Oral script: the 10-minute defence

Museum Guide project, MSA-DATB09-03. One section per slide, to read aloud.

How to use it

- About 1269 words: at a calm pace (about 130 words per minute), this is about 10 minutes. Without the 68 optional words, about 9 minutes. The rest of the 10 minutes is a safety margin for pauses, clicks and nerves.
- Every slide ends with a green TRANSITION box, always built the same way: "We saw that... So now...". Say it while you click to the next slide, and never skip it, even when you are late: it turns the six steps into one story.
- Read it aloud 3 times, with the slides, and time yourself once. Speak slowly; pause after each bold sentence: it is the one the jury must remember.
- Blue text in brackets is a stage cue (where to point); do not read it. Grey sentences marked (optional) can be skipped if you are late.
- If you present with a teammate, a natural split is slides 1 to 5, then slides 6 to 11.

The story, transition by transition

1 to 2: To answer it, we did not jump straight to a model. So now, here is our path: six steps.

2 to 3: So now, let's start with step one: the data, and how we made the test fair.

3 to 4: We saw that our data is ready and our test is fair. So now, step two: we can build our first model.

4 to 5: We saw that our MLP is overfitting. So now, step three: we try to fix this overfitting.

5 to 6: We saw that the overfitting is under control. But our MLP still sees colours, not shapes. So now, step four: we need a network that sees shapes.

6 to 7: We saw how ResNet18 works. So now, let's see what it gives on our paintings.

7 to 8: We saw that ResNet18 reaches 88 percent. But to answer the director, we also need a challenger without any neural network. So now, step five: we build this classical challenger, then we test every model.

8 to 9: We saw how our challenger works. So now, the final test: every model, on 220 paintings never seen before.

9 to 10: We saw that the pretrained network is the most accurate. But accuracy is only half of the director's question. So now, step six: what does it cost on a kiosk?

10 to 11: We saw that it is accurate, fast and cheap. So now, we have everything we need to answer the director.

Slide 1 · Is a neural network worth it on a museum kiosk?

0:00 to 0:45 · 45 s · 98 words

Good morning. Imagine you point your phone at a painting in a museum, and a guide tells you its style. Our museum wants exactly this, on kiosks in its galleries.

We worked on five styles: Baroque, Cubism, Impressionism, Pop Art and Ukiyo-e. *[point at the five paintings]*

The kiosks have small CPUs, the team is small and the budget is limited. So the director asks one question: **is a neural network worth it in production, or would a simpler classical model do the job?**

> **TRANSITION TO SLIDE 2**
> To answer it, we did not jump straight to a model. **So now, here is our path: six steps.**

Slide 2 · Six steps to answer the director

0:45 to 1:20 · 35 s · 80 words

[point at the six steps, one by one] Step one: the data, and a fair test. Step two: our own neural network. Step three: we monitor its training and fix it. Step four: a pretrained ResNet18. Step five: a final test against a classical model. Step six: the cost on a kiosk.

Each step told us what to try next. **At the end of these six steps, we will have our answer.**

> **TRANSITION TO SLIDE 3**
> **So now, let's start with step one: the data, and how we made the test fair.**

Slide 3 · A fair test: one painter, one part

1:20 to 2:15 · 55 s · 124 words

We took 1 500 paintings from WikiArt, 300 per style.

We cut them into three piles: 1 061 to train, 219 to validate, that is, to choose our settings, and 220 to test, used only once, at the very end.

One rule matters a lot: **all the paintings of a known painter go to the same pile.** *[point at the zero]* With a random split, 23 of our 32 known painters would be in two piles, and a model could learn to recognise Monet instead of Impressionism. With our split: zero. *(optional) We measured it: when the model has seen the painter, it scores 11 points higher.*

> **TRANSITION TO SLIDE 4**
> We saw that our data is ready and our test is fair. **So now, step two: we can build our first model.**

Slide 4 · Our MLP learns the paintings by heart

2:15 to 3:20 · 65 s · 137 words

Our first model is the simplest neural network of the course: a multi-layer perceptron built from nn.Linear layers and ReLU. *(optional) It reads the pixels of a 64 by 64 image: 12 288 inputs and 6.4 million parameters.*

Before trusting it, we ran two checks. First, a model that knows nothing should have a loss of ln 5, about 1.61. Ours started at 1.99, too confident, so we made its last layer smaller and got 1.59. Second, it must be able to learn one small batch by heart: it does, 100 percent.

Then the real training. *[point at the curves]* The training accuracy goes to 99.7 percent, but the validation accuracy stays around 40 percent. **This is overfitting: the network learns the paintings by heart.**

> **TRANSITION TO SLIDE 5**
> We saw that our MLP is overfitting. **So now, step three: we try to fix this overfitting.**

Slide 5 · Healthy training, but the MLP overfits

3:20 to 4:20 · 60 s · 129 words

Before fixing it, we looked inside the network to check that the training itself was healthy: the gradient reaches every layer and only 4 percent of the neurons are dead. So the problem really is overfitting.

Then we attacked the overfitting with the tools of session 6, 3 seeds each. *[point at the chart]* All together, weight decay, dropout, batch norm, flips and early stopping take the MLP from 38.8 to 45.5 percent, and the gap between train and validation from 0.61 to 0.12.

But one number shows a deeper limit: **the MLP finds only 7 percent of the Cubist paintings. It sees colours, not shapes.**

> **TRANSITION TO SLIDE 6**
> We saw that the overfitting is under control. But our MLP still sees colours, not shapes.
> **So now, step four: we need a network that sees shapes.**

Slide 6 · ResNet18: a network made for images

4:20 to 5:25 · 65 s · 142 words

There is a type of network made for images, which recognises shapes: the convolutional network. We used ResNet18, the smallest ResNet.

[point at the dark box] Its 18 layers slide small filters over the picture: the first layers find edges and colours, the next ones textures, the last ones shapes. Our MLP saw 12 288 separate pixels.

ResNet18 already learned all this on 1.2 million ImageNet photos. We only have 1 061 paintings, so we keep it as it is: it is frozen. *[point at the 512 numbers]* It turns each painting into 512 numbers. *[point at the blue box]* We remove its last layer, made for ImageNet, and add our own: from these 512 numbers to our 5 styles, 2 565 weights, the only ones we train. **Like hiring an experienced art expert, and only teaching them our five style names.**

> **TRANSITION TO SLIDE 7**
> We saw how ResNet18 works. **So now, let's see what it gives on our paintings.**

Slide 7 · A pretrained ResNet18 sees shapes

5:25 to 6:30 · 65 s · 145 words

[point at the blue bar] Training this last layer takes less than a second, and gives **88.1 percent on validation, almost twice our best MLP.** *[point at the top card]* And it does see shapes: it finds 71 percent of the Cubist paintings, against 7 for our MLP.

We also tried fine-tuning: we unfreeze the last block of ResNet18, so that it also learns from our paintings, 8.4 million weights instead of 2 565. It was less convincing than keeping it frozen: 85.4 percent instead of 88.1. *(optional) With only 1 061 paintings, it starts to learn them by heart.*

[point at the bottom card] We monitored it like the MLP: the gap between train and validation is only 0.09, against 0.61 for our first MLP.

> **TRANSITION TO SLIDE 8**
> We saw that ResNet18 reaches 88 percent. But to answer the director, we also need a challenger without any neural network. **So now, step five: we build this classical challenger, then we test every model.**

Slide 8 · Our classical challenger: a random forest

6:30 to 7:20 · 50 s · 112 words

A classical model cannot read raw pixels well. So we first describe each painting with 102 hand-made numbers: *[point at the second box]* 82 for its colours, and 20 for the direction and strength of its brush strokes.

[point at the tree] Then a random forest: 500 decision trees. Each tree asks yes or no questions on these numbers, like "many dark colours?", until it names a style. Each one learns from a random draw of the paintings, so they make different mistakes. **They vote, and the vote cancels the mistakes.**

[point at the right card] Like every model, we chose its settings on validation: 63.5 percent.

> **TRANSITION TO SLIDE 9**
> We saw how our challenger works. **So now, the final test: every model, on 220 paintings never seen before.**

Slide 9 · Final test: the pretrained network wins

7:20 to 8:15 · 55 s · 118 words

We opened the 220 test paintings only once, for every model. *[point at the top bars]* The pretrained models are at the top, from 83 to 88 percent. The random forest reaches 67; our MLP 41.

Is the gap real? *[point at the blue card]* A paired bootstrap on the same test paintings gives our model a lead of **16.4 points over the random forest, between 9 and 24 points: it is not luck.**

[point at the grey card] *(optional) One honest note: the fine-tuned model is first here, but it was second on validation, and we keep the model chosen before the test.*

> **TRANSITION TO SLIDE 10**
> We saw that the pretrained network is the most accurate. But accuracy is only half of the director's question. **So now, step six: what does it cost on a kiosk?**

Slide 10 · Fast and cheap enough for a kiosk

8:15 to 8:50 · 35 s · 71 words

We timed one photo at a time, from the image to the answer. *[point at the blue card]* The ResNet needs **53 milliseconds on one CPU thread**, 26 with four threads: faster than a blink. Even on a kiosk four times slower, it answers in a fifth of a second.

The model file is 45 megabytes.

> **TRANSITION TO SLIDE 11**
> We saw that it is accurate, fast and cheap. **So now, we have everything we need to answer the director.**

Slide 11 · Our choice: frozen ResNet18 + linear head

8:50 to 9:40 · 50 s · 113 words

[point at the line under the title] On the test, our own MLP reached 41 percent, the best classical model 67, and the frozen ResNet18 83.

So, our answer to the director: yes, a neural network is worth it, but a pretrained one that we reuse. **We recommend a frozen ResNet18 with a linear head.**

[point at the left column] Why? It is accurate: one visitor in six gets a wrong style, against one in three. It is fast enough for a modest CPU, and cheap for a small team: one small layer, retrained in seconds.

[point at the right column] Before the launch, we must test real photos taken at the kiosks, add more styles, and measure the real hardware.

Thank you. We are happy to take your questions.